

METHANE EMISSION MONITORING USING IoT AND BLOCKCHAIN

*A project report Submitted in partial fulfillment of the requirements for the
award of degree of*

BACHELOR OF TECHNOLOGY

in

ELECTRONICS AND COMMUNICATION ENGINEERING

by

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

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2022-2023

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Bonafide Certificate

This is to certify that this project report entitled **“METHANE EMISSION MONITORING USING IoT AND BLOCKCHAIN”** submitted to **Department of Electronics and Communication Engineering, K L Deemed to be University, Guntur**, in connection with the University internship program is a bonafide record of work done by **“MADINENI YUVARAJ UJWAL”** under my supervision at the **“PENTAGON RUGGED SYSTEMS”** from “18-07-2022” to “18-11-2022”.

Signature of the Company Guide with Seal & Designation

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Certificate

This is Certified that the project entitled “**METHANE EMISSION MONITORING USING IoT AND BLOCKCHAIN**” which is a experimental &/ theoretical &/ Simulation&/ hardware work carried out by **MADINENI YUVARAJ UJWAL** (190040664), in partial fulfillment for the award of the degree of **Bachelor of Technology** in Department **Electronics and Communication Engineering**, during the year **2022-23**. The project has been approved as it satisfies the academic requirements.

Department Coordinator

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ACKNOWLEDGEMENT

I have taken efforts in this project. However, it would not have been possible without the kind, support and help of **Dr. S. Manikandan**, Company Guide.

I am highly indebted to **Dr. S. Manikandan**, Project Coordinator for his guidance and constant supervision as well as for providing necessary information regarding the project & also for his support for completing the project.

I would like to express my special gratitude and thanks to all the faculty and staff for giving me such attention and time.

I would like to express our gratitude towards our parents for their kind co-operation and encouragement which help us in completion of this project.

M. YUVARAJ UJWAL

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1.0 ABSTRACT

Global warming is likely to have unimaginable disruptions in every aspect of human life if some actions are not taken immediately to reduce its fast acceleration. Since the Industrial Revolution (1900s), the global mean surface temperature has warmed up approximately +0.8 °C (1.4 °F) due to past and ongoing human-induced emissions of greenhouse gasses as a result of burning of fossil fuels. Scientists and environmentalists warn that if nothing is done in the next decade to deal with climate changes, the global warming may accelerate to around 1.5 °C by 2050, which may lead to climate changes with catastrophic consequences. Carbon Dioxide (CO₂) and Methane (CH₄) are the main contributors for global warming. Blockchain technology can be used to monitor the emissions of greenhouse gases breaking the myth that it is only confined to cryptocurrencies.

Keywords:

Global Warming, Methane, Blockchain, Carbon Dioxide

2.0 INTRODUCTION

Water vapour, carbon dioxide, methane and nitrous oxide are the main greenhouse gases in the atmosphere that trap heat close to Earth's surface. If they were not present, the oceans would freeze, and the planet would not be habitable. But greenhouse gases are so good at trapping heat from the Sun that if their levels rise just a little, Earth's temperature also rises, with potentially significant negative effects for life on Earth.

Various natural processes, such as photosynthesis and respiration for carbon dioxide, or the production of methane by microorganisms in wetlands, control the natural presence of these greenhouse gases in the atmosphere. However, human activity, especially since industrial times, has rapidly increased the amount of some greenhouse gases in the atmosphere, resulting in the Earth system heating up at an unprecedented rate. These anthropogenic greenhouse gases can remain in the atmosphere for years, decades or even longer (depending on the gas) after their emission.

In the present day, Global Warming is a serious topic. It results in the heating up of earth surface, gradually over time due to release of greenhouse gases. Out of all greenhouse gases, Methane (CH₄) and Carbon Dioxide (CO₂) are the main contributors for global warming. Carbon Dioxide lies dormant in the atmosphere for centuries and its reduction doesn't help in reduction of global warming. On the other hand, Methane lies dormant only for a decade, taking reduction measures can be of great aid to reduce global warming. In this report, we are monitoring methane emissions using blockchain technology.

Blockchain is not confined to cryptocurrencies and can help reduce current levels of human-produced carbon dioxide and methane resulting from activities involving factory farming, livestock farming, and deforestation which are releasing huge amounts of greenhouse gases that suffocate the atmosphere.

The many processes controlling the carbon and methane budgets are explained below.

3.0 TECHNOLOGICAL ANALYSIS

The process of methane monitoring in this report uses two main technologies to monitor the data.

They are:

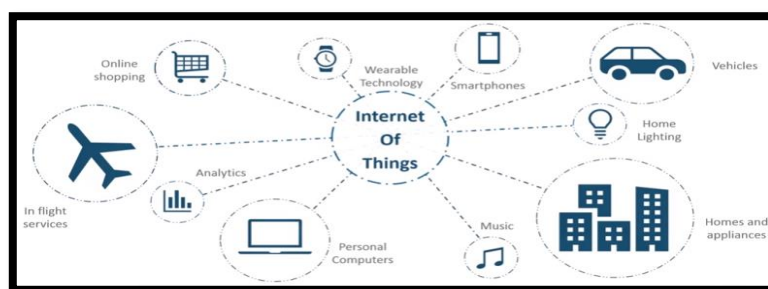
- Internet of Things (IoT)
- Blockchain Technology

3.0.1 Internet of Things (IoT):

The term Internet of Things (IoT) describes about group of things put together that share information among them by connecting with each other through a network. These things connected are mainly devices with sensors or processing abilities that either record and send data or receive the data and process the required information.

IoT is used in different sectors, such as:

- Climate and Environment
- Medical
- Agricultural
- Industrial
- Military
- Energy Management



And many more sectors use the technology of IoT.

The IoT technology is of great demand nowadays and will be in the future in every sector and every application. In this report, methane emission monitoring falls under climate and environment sector. Connecting everyday things embedded with electronics, software, and sensors to internet enabling to collect and exchange data without human interaction called as the Internet of Things (IoT). The term "Things" in the Internet of Things refers to

anything and everything in day to day life which is accessed or connected through the internet. IoT is an advanced automation and analytics system which deals with artificial intelligence, sensor, networking, electronic, cloud messaging etc. to deliver complete systems for the product or services. The system created by IoT has greater transparency, control, and performance.

3.0.2 Blockchain Technology:

The term 'blockchain' refers to a distributed chain of blocks. In other words, it's a distributed database of records or blocks.

In blockchain, the database information can be shared and identified, but not edited.

Blockchain Technology is:

- **Distributed:** The distributed ledger is shared and updated with every incoming data among the nodes connected to the Blockchain. All this is done in real-time as there is no central server controlling the data.
- **Secure:** There is no unauthorized access to blockchain. It is made possible through Permissions and Cryptography.
- **Transparent:** Because every node or participant in Blockchain has a copy of the Blockchain data, they have access to all transaction data. They themselves can verify the identities without the need for mediators.
- **Consensus-Based:** All relevant network participants must agree that a transaction is valid. This is achieved through the use of consensus algorithms.
- **Flexible:** Smart Contracts which are executed based on certain conditions can be written into the platform. Blockchain Network can evolve in pace with business processes.

Benefits of Blockchain Technology are:

- **Time-Saving:** No central authoritative verification needed for processing, making the process faster and cheaper.
- **Cost-Saving:** A blockchain network reduces expenses in ways such as, no need for third-party verification. Participants can share data directly. Intermediaries are reduced. Data sharing is minimized as every participant has a copy of shared ledger.

- **Tighter Security:** No one can tamper with blockchain data as it is shared among millions of participants. The system is safe against errors and hacking.

3.0.3 ARDUINO IDE:

Arduino is an open-source electronics platform based on easy-to-use hardware and software. Arduino boards are able to read inputs - light on a sensor, a finger on a button, or a Twitter message - and turn it into an output - activating a motor, turning on an LED, publishing something online. You can tell your board what to do by sending a set of instructions to the microcontroller on the board. To do so you use the Arduino programming language (based on Wiring), and the Arduino Software (IDE), based on Processing.

Over the years Arduino has been the brain of thousands of projects, from everyday objects to complex scientific instruments. A worldwide community of makers - students, hobbyists, artists, programmers, and professionals - has gathered around this open-source platform, their contributions have added up to an incredible amount of accessible knowledge that can be of great help to novices and experts alike.

Arduino was born at the Ivrea Interaction Design Institute as an easy tool for fast prototyping, aimed at students without a background in electronics and programming. As soon as it reached a wider community, the Arduino board started changing to adapt to new needs and challenges, differentiating its offer from simple 8-bit boards to products for IoT applications, wearable, 3D printing, and embedded environments. All Arduino boards are completely open-source, empowering users to build them independently and eventually adapt them to their particular needs. The software, too, is open-source, and it is growing through the contributions of users worldwide.

WHY ARDUINO?

Thanks to its simple and accessible user experience, Arduino has been used in thousands of different projects and applications. The Arduino software is easy-to-use for beginners, yet flexible enough for advanced users. It runs on Mac, Windows, and Linux. Teachers and students use it to build low cost scientific instruments, to prove chemistry and physics principles, or to get started with programming and robotics. Designers and architects build interactive prototypes, musicians and artists use it for installations and to experiment with new musical instruments. Makers, of course, use it to build many of the projects exhibited at the Maker Faire, for example. Arduino is a key tool to learn new things. Anyone - children, hobbyists, artists, programmers - can start tinkering just following the step by step instructions of a kit, or sharing ideas online with other members of the Arduino community.

There are many other microcontrollers and microcontroller platforms available for physical computing. Parallax Basic Stamp, Netmedia's BX-24, Phidgets, MIT's Handyboard, and many others offer similar functionality. All of these tools take the messy details of microcontroller programming and wrap it up in an easy-to-use package. Arduino also simplifies the process of working with microcontrollers, but it offers some advantage for teachers, students, and interested amateurs over other systems:

- Inexpensive - Arduino boards are relatively inexpensive compared to other microcontroller platforms. The least expensive version of the Arduino module can be assembled by hand, and even the pre-assembled Arduino modules cost less than \$50
- Cross-platform - The Arduino Software (IDE) runs on Windows, Macintosh OSX, and Linux operating systems. Most microcontroller systems are limited to Windows.
- Simple, clear programming environment - The Arduino Software (IDE) is easy-to-use for beginners, yet flexible enough for advanced users to take advantage of as well. For teachers, it's conveniently based on the Processing programming environment, so students learning to program in that environment will be familiar with how the Arduino IDE works.
- Open source and extensible software - The Arduino software is published as open source tools, available for extension by experienced programmers. The language can be expanded through C++ libraries, and people wanting to understand the technical details can make the

leap from Arduino to the AVR C programming language on which it's based. Similarly, you can add AVR-C code directly into your Arduino programs if you want to.

- Open source and extensible hardware - The plans of the Arduino boards are published under a Creative Commons license, so experienced circuit designers can make their own version of the module, extending it and improving it. Even relatively inexperienced users can build the breadboard version of the module in order to understand how it works and save money.



3.0.4 MQ4 SENSOR:

MQ Sensors are Gas Sensors. Gas sensors (also known as gas detectors) are electronic devices that detect and identify different types of gasses. They are commonly used to detect toxic or explosive gasses and measure gas concentration. Gas sensors are employed in factories and manufacturing facilities to identify gas leaks, and to detect smoke and carbon monoxide in homes.

Gas sensors vary widely in size (portable and fixed), range, and sensing ability. They are often part of a larger embedded system, such as security systems, and they are normally connected to an audible alarm or interface. Because gas sensors are constantly interacting with air and other gasses, they have to be calibrated more often than many other types of sensors.

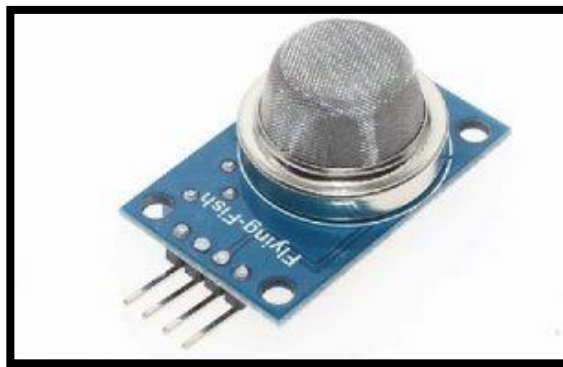
MQ2 is one of the commonly used gas sensors in MQ sensor series. It is a Metal Oxide Semiconductor (MOS) type Gas Sensor also known as Chemiresistors as the detection is based upon change of resistance of the sensing material when the Gas comes in contact with the material. Using a simple voltage divider network, concentrations of gas can be detected.

Some MQ series of gas sensors are:

- MQ-2 - Methane, Butane, LPG, smoke
- MQ-3 - Alcohol, Ethanol, smoke
- MQ-4 - Methane, CNG Gas
- MQ-5 - Natural gas, LPG

- MQ-6 - LPG, butane gas
- MQ-7 - Carbon Monoxide
- MQ-8 - Hydrogen Gas
- MQ-9 - Carbon Monoxide, flammable gasses
- MQ131 - Ozone
- MQ135 - Air Quality (CO, Ammonia, Benzene, Alcohol, smoke)
- MQ136 - Hydrogen Sulfide gas
- MQ137 - Ammonia
- MQ138 - Benzene, Toluene, Alcohol, Acetone, Propane, Formaldehyde gas, Hydrogen
- MQ214 - Methane, Natural gas

In this project, we use MQ4 Sensor. MQ4 methane gas sensor is a MOS (metal oxide semiconductor) type sensor, used to detect the methane gas concentration within the air at either home or industries & generates output like analog voltage by reading it. Here, the range of concentration for sensing ranges from 300 pm – 10,000 ppm which is appropriate for the detection of a leak.



MQ4 Methane Gas Sensor

The MQ4 sensor contains four pins. They are:

- **Voltage (VCC) Pin:** This pin provides voltage to the sensor. The typical operating voltage is +5V.
- **Ground (GND) Pin:** This pin grounds the module to the ground terminal of the system.
- **D0 Pin (Digital Out):** This pin provides the digital output based on a threshold value with a potentiometer

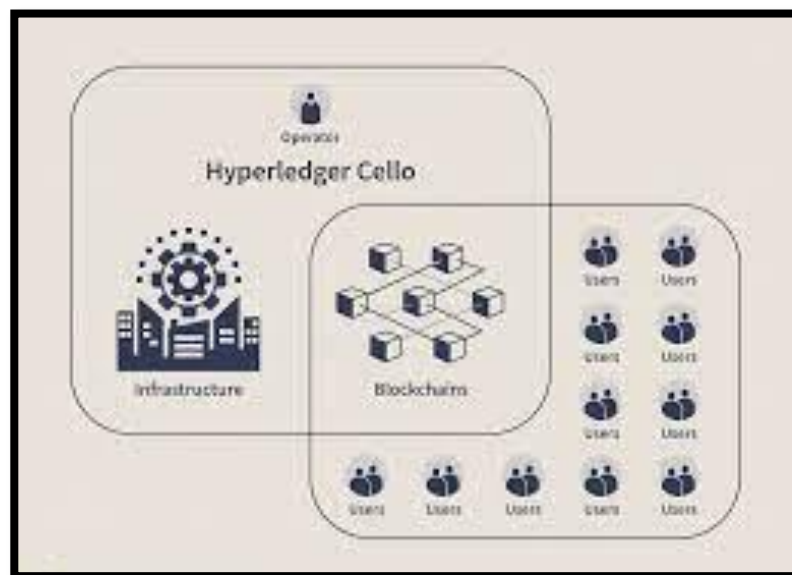
- **A0 Pin (Analog Out):** This pin provides output analog voltage which ranges from 0 to 5V based on the intensity of the gas.

3.0.5 CLOUD INTERFACE WITH BLOCKCHAIN:

Blockchain-as-a-service (BaaS) is a cloud-managed platform that allows people and organizations to more easily and quickly build, host, and manage blockchain applications. With blockchain cloud services, transactional data from multiple sources can be easily collected, integrated, and shared.

The collected data can be verified in the blockchain database with consensus mechanism. After verification, the data can be hashed to the blockchain and record it in the blockchain cloud.

Popular example for Blockchain cloud is: Ethereum (used in cryptocurrencies).



Blockchain-as-a-service (BaaS)

4.0 COMPONENTS REQUIRED:

4.0.1 ARDUINO UNO:

The Arduino Uno is a microcontroller board based on the ATmega328 . It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header, and a reset button. It

contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started.

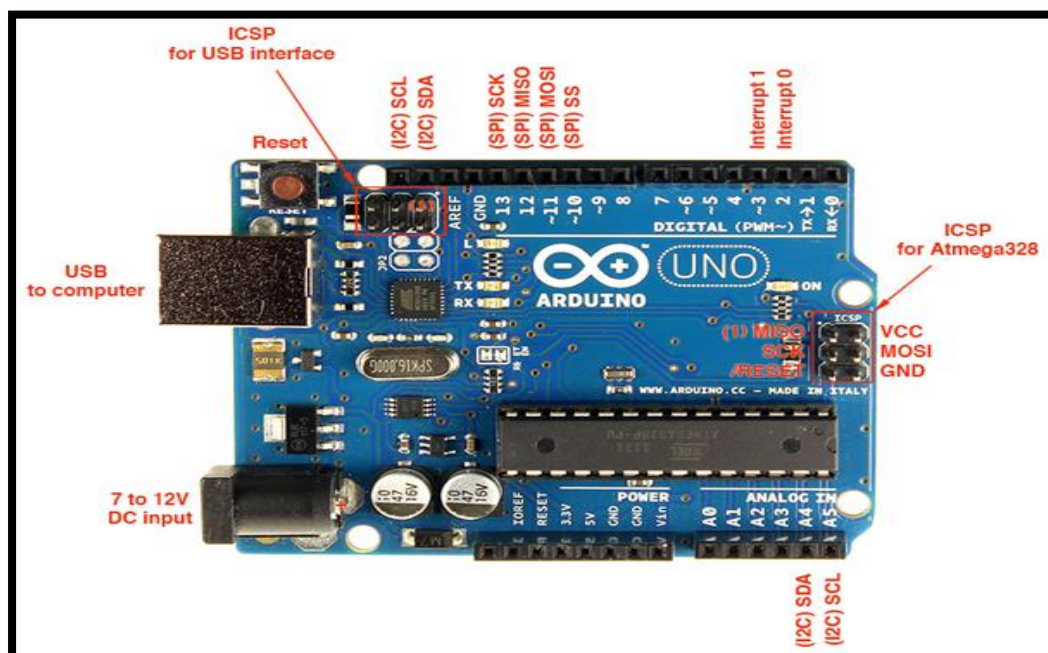
The Uno differs from all preceding boards in that it does not use the FTDI USB-to-serial driver chip. Instead, it features the Atmega16U2 (Atmega8U2 up to version R2) programmed as a USB-to-serial converter. The Uno board has a resistor pulling the 8U2 HWB line to ground, making it easier to put into DFU mode.

➤ The board has the following new features:

- 1.0 pinout: added SDA and SCL pins that are near to the AREF pin and two other new pins placed near to the RESET pin, the IOREF that allow the shields to adapt to the voltage provided from the board. In future, shields will be compatible with both the board that uses the AVR, which operates with 5V and with the Arduino Due that operates with 3.3V. The second one is a not connected pin, that is reserved for future purposes.
- Stronger RESET circuit.
- ATmega 16U2 replace the 8U2.

"Uno" means one in Italian and is named to mark the upcoming release of Arduino 1.0. The Uno and version 1.0 will be the reference versions of Arduino, moving forward. The Uno is the latest in a series of USB Arduino boards.

Arduino pin Diagram:



POWER:

The Arduino Uno can be powered via the USB connection or with an external power supply. The power source is selected automatically.

External (non-USB) power can come either from an AC-to-DC adapter (wall-wart) or battery. The adapter can be connected by plugging a 2.1mm center-positive plug into the board's power jack. Leads from a battery can be inserted in the Gnd and Vin pin headers of the POWER connector.

The board can operate on an external supply of 6 to 20 volts. If supplied with less than 7V, however, the 5V pin may supply less than five volts and the board may be unstable. If using more than 12V, the voltage regulator may overheat and damage the board. The recommended range is 7 to 12 volts.

The power pins are as follows:

- **VIN.** The input voltage to the Arduino board when it's using an external power source (as opposed to 5 volts from the USB connection or other regulated power source). You can supply voltage through this pin, or, if supplying voltage via the power jack, access it through this pin.
- **5V.** This pin outputs a regulated 5V from the regulator on the board. The board can be supplied with power either from the DC power jack (7 - 12V), the USB connector (5V), or the VIN pin of the board (7-12V). Supplying voltage via the 5V or 3.3V pins bypasses the regulator, and can damage your board. We don't advise it.
- **3V3.** A 3.3 volt supply generated by the on-board regulator. Maximum current draw is 50 mA.
- **GND.** Ground pins.
- **IOREF.** This pin on the Arduino board provides the voltage reference with which the microcontroller operates. A properly configured shield can read the IOREF pin voltage and select the appropriate power source or enable voltage translators on the outputs for working with the 5V or 3.3V.

MEMORY:

The ATmega328 has 32 KB (with 0.5 KB used for the bootloader). It also has 2 KB of SRAM and 1 KB of EEPROM (which can be read and written with the [EEPROM library](#)).

Input and Output:

Each of the 14 digital pins on the Uno can be used as an input or output, using `pinmode`, `digitalwrite`, and `digitalread` functions. They operate at 5 volts. Each pin can provide or receive a maximum of 40 mA and has an internal pull-up resistor (disconnected by default) of 20-50 kOhms. In addition, some pins have specialized functions:

- **Serial: 0 (RX) and 1 (TX).** Used to receive (RX) and transmit (TX) TTL serial data. These pins are connected to the corresponding pins of the ATmega8U2 USB-to-TTL Serial chip.
- **External Interrupts: 2 and 3.** These pins can be configured to trigger an interrupt on a low value, a rising or falling edge, or a change in value. See the [attachInterrupt\(\)](#) function for details.
- **PWM: 3, 5, 6, 9, 10, and 11.** Provide 8-bit PWM output with the [analogWrite\(\)](#) function.
- **SPI: 10 (SS), 11 (MOSI), 12 (MISO), 13 (SCK).** These pins support SPI communication using the [SPI library](#).
- **LED: 13.** There is a built-in LED connected to digital pin 13. When the pin is HIGH value, the LED is on, when the pin is LOW, it's off.

The Uno has 6 analog inputs, labeled A0 through A5, each of which provide 10 bits of resolution (i.e. 1024 different values). By default they measure from ground to 5 volts, though it is possible to change the upper end of their range using the AREF pin and the [analogReference](#) function. Additionally, some pins have specialized functionality:

- **TWI: A4 or SDA pin and A5 or SCL pin.** Support TWI communication using the [Wire library](#).

There are a couple of other pins on the board:

- **AREF.** Reference voltage for the analog inputs. Used with [analogReference](#).

- **Reset.** Bring this line LOW to reset the microcontroller. Typically used to add a reset button to shields which block the one on the board.

See also the mapping between Arduino pins and ATmega328 ports. The mapping for the Atmega8, 168, and 328 is identical.

Communication:

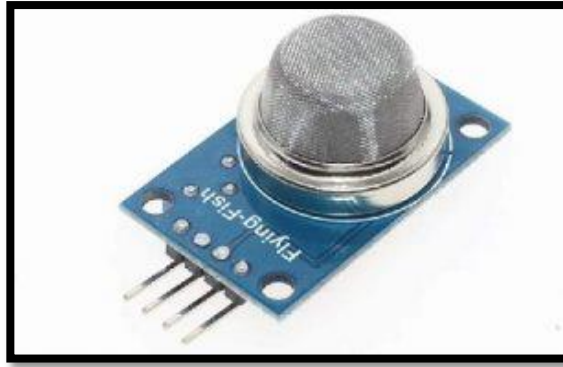
The Arduino Uno has a number of facilities for communicating with a computer, another Arduino, or other microcontrollers. The ATmega328 provides UART TTL (5V) serial communication, which is available on digital pins 0 (RX) and 1 (TX). An ATmega16U2 on the board channels this serial communication over USB and appears as a virtual com port to software on the computer. The '16U2 firmware uses the standard USB COM drivers, and no external driver is needed. However, on Windows, a .inf file is required. The Arduino software includes a serial monitor which allows simple textual data to be sent to and from the Arduino board. The RX and TX LEDs on the board will flash when data is being transmitted via the USB-to-serial chip and USB connection to the computer (but not for serial communication on pins 0 and 1).

USB Overcurrent Protection:

The Arduino Uno has a resettable polyfuse that protects your computer's USB ports from shorts and overcurrent. Although most computers provide their own internal protection, the fuse provides an extra layer of protection. If more than 500 mA is applied to the USB port, the fuse will automatically break the connection until the short or overload is removed.

4.0.2 MQ4 SENSOR:

MQ4 methane gas sensor is a MOS (metal oxide semiconductor) type sensor, used to detect the methane gas concentration within the air at either home or industries & generates output like analog voltage by reading it. Here, the range of concentration for sensing ranges from 300 pm – 10,000 ppm which is appropriate for the detection of a leak.



MQ4 Methane Gas Sensor

4.0.3 LCD DISPLAY:

LCD (Liquid Crystal Display) is a type of flat panel display which uses liquid crystals in its primary form of operation. LEDs have a large and varying set of use cases for consumers and businesses, as they can be commonly found in smartphones, televisions, computer monitors and instrument panels.

LCDs were a big leap in terms of the technology they replaced, which include light-emitting diode (LED) and gas-plasma displays. LCDs allowed displays to be much thinner than cathode ray tube (CRT) technology. LCDs consume much less power than LED and gas-display displays because they work on the principle of blocking light rather than emitting it. Where an LED emits light, the liquid crystals in an LCD produces an image using a backlight.

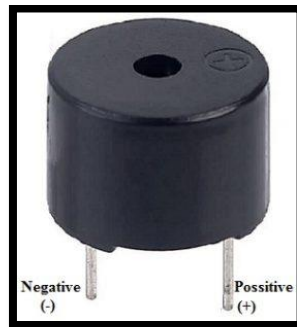
As LCDs have replaced older display technologies, LCDs have begun being replaced by new display technologies such as OLEDs.



4.0.4 BUZZER:

An audio signaling device like a beeper or buzzer may be electromechanical or piezoelectric or mechanical type. The main function of this is to convert the signal from audio to sound. Generally, it is powered through DC voltage and used in timers, alarm

devices, printers, alarms, computers, etc. Based on the various designs, it can generate different sounds like alarm, music, bell & siren.



4.0.5 JUMPER WIRES:

A **jump wire** (also known as **jumper**, **jumper wire**, **DuPont wire**) is an electrical wire, or group of them in a cable, with a connector or pin at each end (or sometimes without them – simply "tinned"), which is normally used to interconnect the components of a breadboard or other prototype or test circuit, internally or with other equipment or components, without soldering.^[1]

Individual jump wires are fitted by inserting their "end connectors" into the slots provided in a breadboard, the header connector of a circuit board, or a piece of test equipment.



4.0.6 DATA CABLE:

This cable is **used to interface any of the Arduino board with your computer**, you can also connect your USB printer, scanner, and more to your computer. These cables Transmits data at high speeds with the error-free, high-performance transmission.

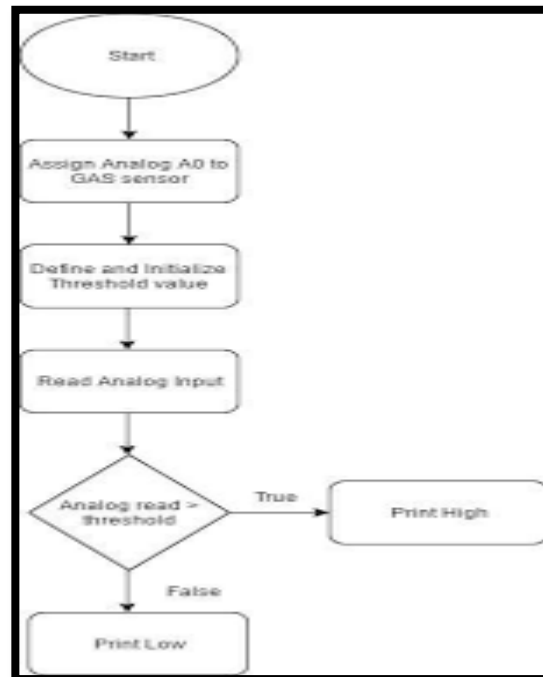


4.0.7 POWER SUPPLY:

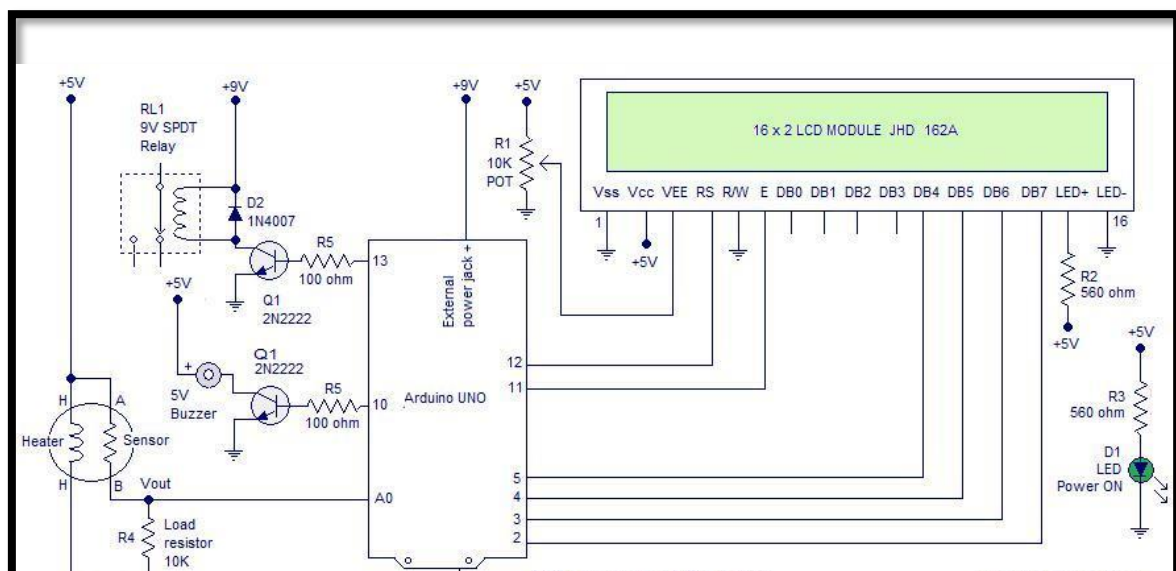
All Arduino boards need electric power to function. A power supply is what is used to provide electric power to the boards and typically can be a battery, USB cable, AC adapter or a regulated power source device. There are different ways to power your Arduino board. The most common way is through the USB connector available on every board, but there are a few other possibilities to power your board.

5.0 FLOWCHART AND CIRCUIT DIAGRAM

5.0.1 FLOW CHART:



5.0.2 CIRCUIT DIAGRAM:



6.0 IMPLEMENTATION:

The project is implemented in the following steps:

STEP1: Research

It deals with the research on methane emissions and its causes along with the statistics on each type of cause.

STEP2: Users

It deals with identifying the users based on the cause of methane emissions.

STEP3: Technologies

It deals with identifying the technologies that can be used to monitor methane emissions.

STEP4: Learning Technologies

It deals with learning the technologies and their constraints.

STEP5: Hardware

Identifying the hardware required that can hold the required technology of blockchain in it.

STEP6: Interfacing

It deals with interfacing the hardware with the technologies.

STEP7: Circuit Building

Build a circuit with the required components and interfacing the technology and monitoring the detected data, through the cloud.

STEP8: Prototype

Developing a prototype as required for the project.

STEP9: Presentation

Presenting the prototype and explaining its function.

By following the above steps in a sequence, we can develop a prototype and explain its function.

7.0 CODE AND OUTPUT:

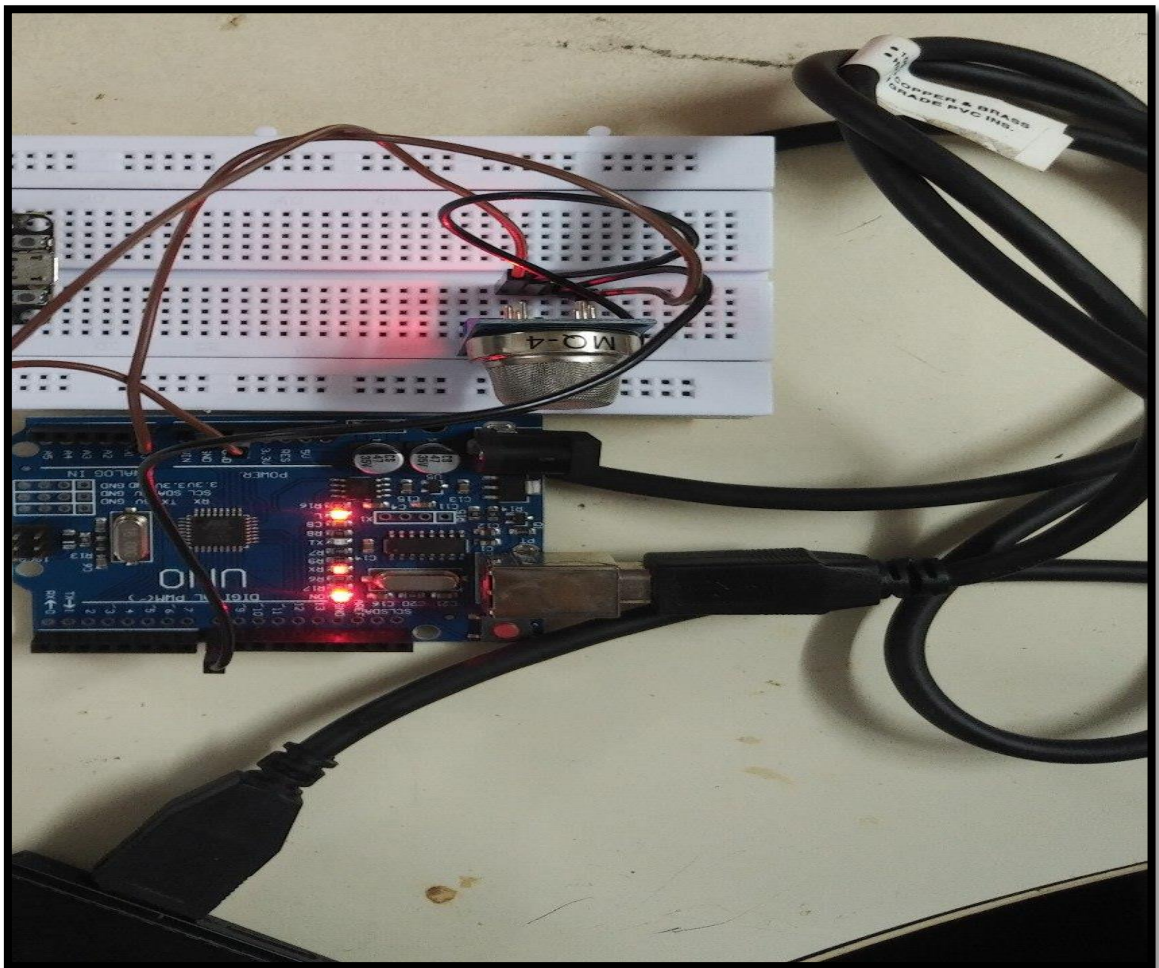
7.0.1 ARDUINO CODE:

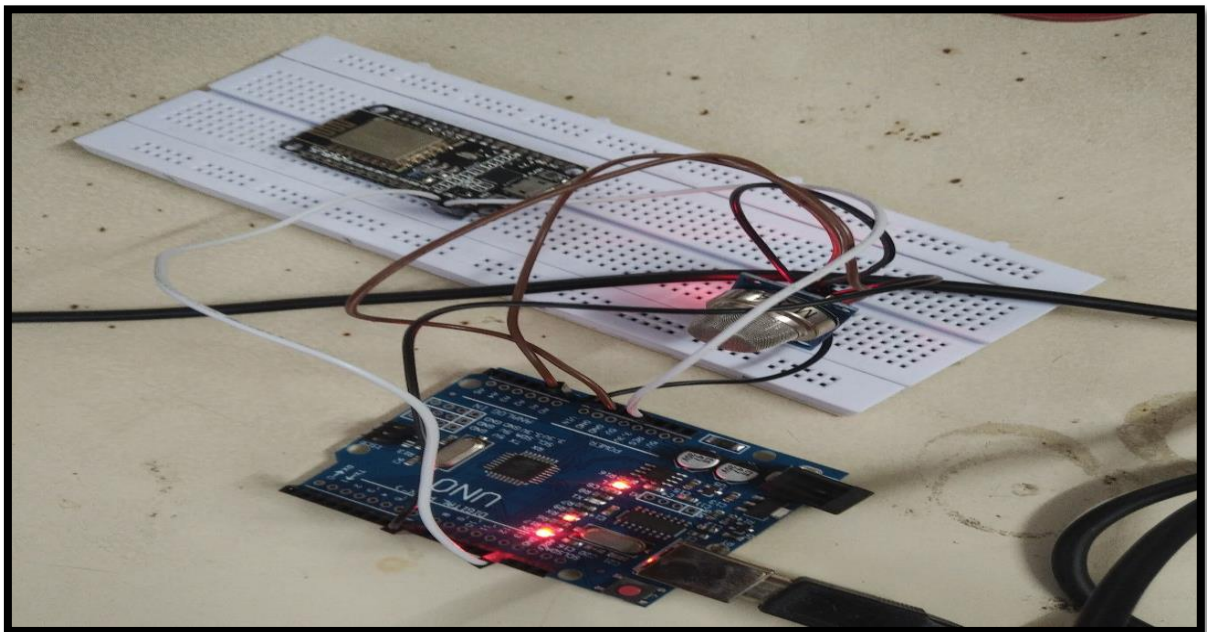
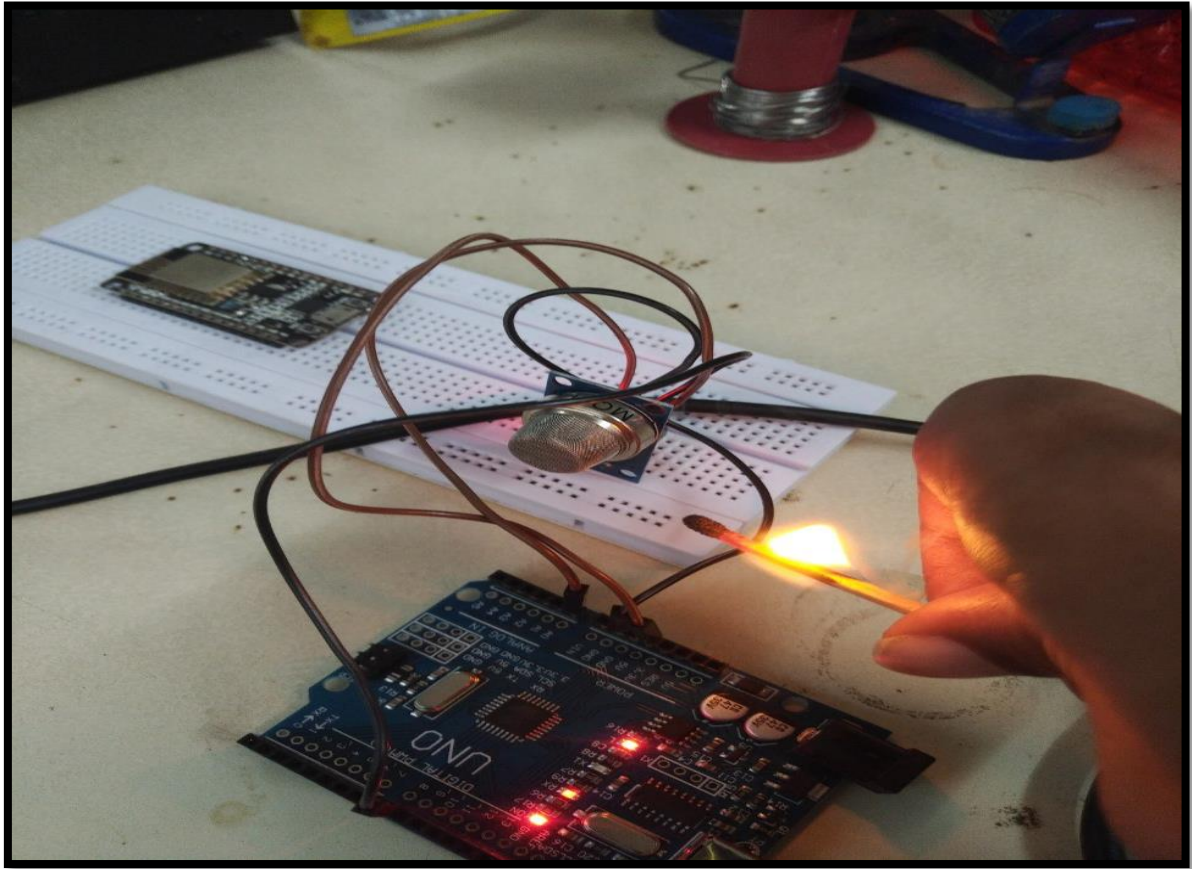
```
#include <ESP8266WiFi.h>
String apiKey = "SKP9YQY2CFVNK919"; // Enter your Write API key from ThingSpeak
const char *ssid = "Leo_Mr_1008"; // replace with your wifi ssid and wpa2 key
const char *pass = "yuvi@w00c m";
const char* server = "api.thingspeak.com";
WiFiClient client;
void setup()
{
  Serial.begin(115200);
  delay(10);
  Serial.println("Connecting to ");
  Serial.println(ssid);
  WiFi.begin(ssid, pass);
  while (WiFi.status() != WL_CONNECTED)
  {
    delay(500);
    Serial.print(".");
  }
  Serial.println("");
  Serial.println("WiFi connected");
}
void loop()
{
  float h = analogRead(A0);
  if (isnan(h))
  {
    Serial.println("Failed to read from MQ-5 sensor!");
    return;
  }

  if (client.connect(server, 80)) // "184.106.153.149" or api.thingspeak.com
  {
    String postStr = apiKey;
    postStr += "&field1=";
    postStr += String(h/1023*100);
    postStr += "\r\n";
    client.print("POST /update HTTP/1.1\r\n");
    client.print("Host: api.thingspeak.com\r\n");
    client.print("Connection: close\r\n");
    client.print("X-THINGSPEAKAPIKEY: " + apiKey + "\r\n");
    client.print("Content-Type: application/x-www-form-urlencoded\r\n");
    client.print("Content-Length: ");
    client.print(postStr.length());
    client.print("\n\n");
```

```
client.print(postStr);  
Serial.print("Gas Level: ");  
Serial.println(h/1023*100);  
Serial.println("Data Send to Thingspeak");  
}  
delay(500);  
client.stop();  
Serial.println("Waiting...");  
  
// thingspeak needs minimum 15 sec delay between updates.  
delay(1500);  
}
```

7.0.2 ASSEMBLY:





7.0.3 OUTPUT:

```
Methane Conentration: 164
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8.0 FUTURE IMPLEMENTATION:

1. This blockchain based methane monitoring can be improvised using Artificial Intelligence (AI).
2. AI tools can detect methane emissions from plume images and can report any changes in the monitoring data.
3. The tools can also suggest respective measures needed to be taken based on the data.
4. Also, AI can train blockchain applications such that the process of validating in blockchain becomes much simpler.
5. Thus, interfacing AI to this blockchain project is the future implementation of the project.

9.0 CONCLUSION:

Hence, methane emissions can be monitored using blockchain and respective measures can be taken to reduce global warming to a greater extent. This project can also be used in daily lives such as cattle feed and livestock which is a common methane factor, to reduce the emission by taking necessary action.

10.0 REFERENCES:

1. EPA, “Greenhouse Gas Emissions of Typical Passenger Vehicle”,
<https://www.epa.gov/greenvehicles/greenhouse-gas-emissions-typical-passenger-vehicle>.
2. IoT, Business Intelligence and Blockchain reduce methane emissions in cattle farm.
<https://www.libelium.com/libeliumworld/success-stories/iot-business-intelligence-and-blockchain-technology-reduces-methane-emissions-in-a-cattle-farm/>.
3. Accessing Blockchain on NodeMCU
<https://hackaday.com/2018/05/02/using-blockchain-explorer-apis-on-nodemcu/>